

Discrete-time Optimal Controller for the Stator Field Oriented Induction Motor Drives

Osama S. Ebrahim^{*}, Praveen K. Jain^{*}, *Fellow IEEE*, and Goel Nishith^{**}

Energy and Power Electronics Applied Research Lab^{*}, Queen's University, Kingston, ON, Canada

bayoumyo@post.queensu.ca, praveen.jain@queensu.ca

Cistel Technology^{**}, Ottawa, ON, Canada

nishith@cistel.com

Abstract— In this paper we introduce a new discrete-time vector control scheme for the induction motor (IM) drives. The scheme is based on the optimal control theory with preview. The control system achieves rotor speed control and stator field orientation control via an optimized voltage command. The synthesized controller has good tracking capability and can cure the adverse phenomena caused by parameters variations and modeling mismatch. Furthermore, the preview feedforward control principle is utilized to improve the transient response of the controlled drive. A laboratory prototype has been built to verify the proposed control strategy.

Key Words— Vector control, digital optimal control, Preview Control Induction motor, SVPWM.

I. INTRODUCTION

SINCE 1980's, field oriented control (FOC) has been recognized as the algorithm which gives the induction motor (IM) drive fast dynamic performance. The basic FOC scheme is based on the rotor flux orientation. However, this strategy is sensitive to both the rotor speed and rotor time constant which varies greatly with temperature, skin effect and, the flux saturation level in the iron core. Therefore, errors can be expected in such method. The online tuning methods that have been developed to overcome these difficulties usually add to the system complexity. On the other hand, the stator flux is closely related to the stator terminal quantities and resistance which can be easily measured with good accuracy. Consequently, using the stator flux instead of the rotor flux as a feedback signal leads to implementation of the stator flux oriented control (SFOC) which is more accurate and robust control [1][2]. In the SFOC, the stator leakage inductance is moved to the rotor side [3]. Accordingly, the stator flux response will follow immediately the input voltage command as long as the stator resistance voltage drop is small. Hence, voltage control is more preferred over the stator flux orientation.

In today's industrial world, pulse width modulated voltage source inverters (PWM-VSI) have been widely accepted because of existing low cost, high performance switching devices and needless to sophisticated feedback schemes which are mandatory employed to stabilize the motion of the IM driven by current source inverters (CSI). Among the different PWM techniques for the VSI, the space vector (SVPWM) provides low harmonic distortion in the output voltage, efficient use for the dc-bus voltage, and it is very convenient for digital implementation [4]. Furthermore, advantages of the

SVPWM such as; constant switching frequency, well defined harmonic spectrum, optimum switching pattern, minimization of switching losses and good utilization of the dc-bus voltage can not be satisfied by classical current control strategies [1],[5].

Various modern control theories such as; optimal control, nonlinear control, variable structure control, neural network, etc., have been applied to the IM to improve the drive performance. Constructing a direct field oriented control system for the IM drives based on the state space method have been addressed in [6]-[8]. However, these solutions require accurate information of the different machine parameters and imply stator current control.

To alleviate these difficulties, this paper introduces a new optimal state vector control for the IM drives. A discrete-time model for the IM suitable for voltage control is derived in the synchronous reference frame using vector method. The model describes the machine as 3-input, 3-output controlled object. The three input variables are the synchronous angular speed and the two components of the stator voltage space vector. Whilst the three output variables are the rotor speed and the two components of the stator flux vector. The objective of the synthesized optimal controller is to achieve rotor speed control and orientation of the stator flux at constant magnitude. An augmented error system is constructed to obtain good tracking performance in spite of modeling errors and disturbances. A feedforward preview control loop is implemented using the desired speed and load torque signals to enhance the transient response of the controlled drive. In the following, the proposed control strategy will be described and validated.

II. IM STATE MODEL

The dynamic equations of the 3-phase IM can be expressed in a synchronously rotating (α - β) frame as

$$\frac{d}{dt} \begin{bmatrix} \omega_r \\ \psi_{\alpha s} \\ \psi_{\beta s} \\ i_{\alpha s} \\ i_{\beta s} \end{bmatrix} = \begin{bmatrix} -\frac{F}{J} \omega_r + \frac{3P}{4J} (\psi_{\alpha s} i_{\beta s} - \psi_{\beta s} i_{\alpha s}) - \frac{1}{J} T_L \\ \omega_e \psi_{\beta s} - R_s i_{\alpha s} + v_{\alpha s} \\ -\omega_e \psi_{\alpha s} - R_s i_{\beta s} + v_{\beta s} \\ \frac{\gamma_r}{\sigma L_s} \psi_{\alpha s} + \frac{\omega_r}{\sigma L_s} \psi_{\beta s} - \frac{(\gamma_s + \gamma_r)}{\sigma} i_{\alpha s} + \omega_s i_{\beta s} + \frac{1}{\sigma L_s} v_{\alpha s} \\ -\frac{\omega_r}{\sigma L_s} \psi_{\alpha s} + \frac{\gamma_r}{\sigma L_s} \psi_{\beta s} - \omega_s i_{\alpha s} - \frac{(\gamma_s + \gamma_r)}{\sigma} i_{\beta s} + \frac{1}{\sigma L_s} v_{\beta s} \end{bmatrix} \quad (1)$$

Where $i_{\alpha s}$, $i_{\beta s}$, $\psi_{\alpha s}$, $\psi_{\beta s}$, $v_{\alpha s}$ and $v_{\beta s}$ are α , β components of the stator current, flux linkage and voltage space vectors, respectively. The symbols ω_s , ω_r and ω_s denote the synchronous, rotor, and slip angular speeds, respectively, while P , T_L , F , and J are the total number of poles, load torque, equivalent mechanical friction and inertia constants, respectively. Further, L_m denotes the magnetizing inductance, while L_s , R_s , L_r and R_r are the stator and rotor self inductances and resistances, respectively. The coefficients γ_s , γ_r and σ are specified by

$$\gamma_s = \frac{R_s}{L_s}, \quad \gamma_r = \frac{R_r}{L_r}, \quad \sigma = 1 - \frac{L_m^2}{L_s L_r} \quad (2)$$

Equation (1) can be rewritten as a nonlinear state model as

$$\dot{x}(t) = f(x(t), u(t), d(t)) \quad (3)$$

The state $x(t)$, input $u(t)$, and the disturbance $d(t)$ are given by

$$\begin{aligned} x(t) &= [\omega_r \quad \psi_{\alpha s} \quad \psi_{\beta s} \quad i_{\alpha s} \quad i_{\beta s}]^T \\ u(t) &= [\omega_e \quad v_{\alpha s} \quad v_{\beta s}]^T \\ d(t) &= T_L \end{aligned} \quad (4)$$

The superscript "T" denotes matrix transposing. Linearizing (3) around a nominal operating point (x_0, u_0, d_0) by the small signal approximation technique yields

$$\begin{aligned} \dot{x}(t) &= A_0 x(t) + B_0 u(t) + C_0 d(t) \\ y(t) &= E x(t) = [\omega_r \quad \psi_{\alpha s} \quad \psi_{\beta s}]^T \end{aligned} \quad (5)$$

The constant matrices A_0 , B_0 , C_0 and E_0 are given by

$$\begin{aligned} A_0 &= \left. \frac{\partial f(\cdot)}{\partial x} \right|_{(x_0, u_0, d_0)}, \quad B_0 = \left. \frac{\partial f(\cdot)}{\partial u} \right|_{(x_0, u_0, d_0)}, \\ C_0 &= \left. \frac{\partial f(\cdot)}{\partial d} \right|_{(x_0, u_0, d_0)} \quad \& \quad E = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{aligned} \quad (6)$$

The discrete-time equivalent of (5) is given by [3]

$$\begin{aligned} x(k+1) &= A x(k) + B u(k-1) + C d(k) \\ y(k) &= E x(k) \end{aligned} \quad (7)$$

The control input $u(k)$ is delayed by one sample period to compensate for the microprocessor execution time.

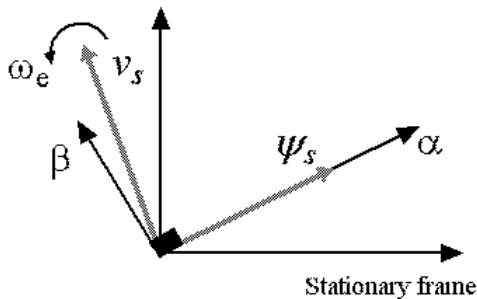


Fig. 1 Orientation of the stator flux vector.

III. CONTROL OBJECTIVES

Basically, the idea of the SFOC is to provide a means of rotor speed control and orientation of the stator flux at constant magnitude. It is clear from (7), our selection for the state, input, and output variables describing the IM model allows realization of such objectives. Where, (7) describes the IM as a 3-input, 3-output controlled object in a synchronous frame rotating at the stator voltage angular frequency as shown in Fig. 1. The synthesized control should adjust the instantaneous values of the three input variables ω_s , $v_{\alpha s}$, and $v_{\beta s}$ arbitrarily to achieve rotor speed control, field orientation control ($\psi_{\beta s}=0$) and constant flux control ($\psi_{\alpha s} = \text{constant}$). Such control input can be translated into equivalent 3-phase voltages on the stator terminals using an industry standard SVPWM-VSI.

IV. CONTROLLER DESIGN

To get output tracking in spite of the modeling errors and disturbances, an error system can be constructed as follows. Define an error signal $e(k)$ as the difference between the desired variable $R(k)$ and the output variable $y(k)$,

$$e(k) = R(k) - y(k) \quad (8)$$

Let us define an augmented state vector as

$$X(k) = [e(k-1) \quad \Delta e(k) \quad \Delta x(k) \quad \Delta u(k-1)] \quad (9)$$

Where Δ is the first difference operator, i.e., $\Delta e(k) = e(k) - e(k-1)$. From (7)-(9), an error system, describing the dynamics of the errors, can be constructed as

$$X(k+1) = \Phi X(k) + \Gamma \Delta u(k) + G_r \Delta Z(k) + G_d \Delta d(k) \quad (10)$$

Here,

$$\begin{aligned} \Phi &= \begin{bmatrix} I_3 & I_3 & 0 & 0 \\ 0 & I_3 & E(I_s - A) & -EB \\ 0 & 0 & A & B \\ 0 & 0 & 0 & 0 \end{bmatrix}, \\ \Gamma &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ I_3 \end{bmatrix}, \quad G_r = \begin{bmatrix} 0 \\ I_3 \\ 0 \\ 0 \end{bmatrix}, \quad G_d = \begin{bmatrix} 0 \\ -EC \\ C \\ 0 \end{bmatrix} \\ \Delta Z(k+1) &= \Delta R(k+1) - \Delta R(k) \end{aligned} \quad (11)$$

and I_n is an identity matrix of dimension n .

A performance index J is selected as

$$J = \sum_{j=0}^{\infty} X(j+1)^T Q X(j+1) + \Delta u(j)^T R \Delta u(j) \quad (12)$$

Minimization of the performance index in (12) subject to the constraints in (10) leads to the following real time optimal preview control law [3], [8]

$$\begin{aligned} u(k) &= g_1 \sum_{i=0}^k e(i) + (g_2 - g_1)e(k) + g_3 x(k) + \\ &g_4 u(k-1) + \sum_{j=1}^M \{F_{rj} [\Delta R(k+j) - \Delta R(j)] \\ &+ F_{dj} [d(k+j-1) - d(j-1)]\} \end{aligned} \quad (13)$$

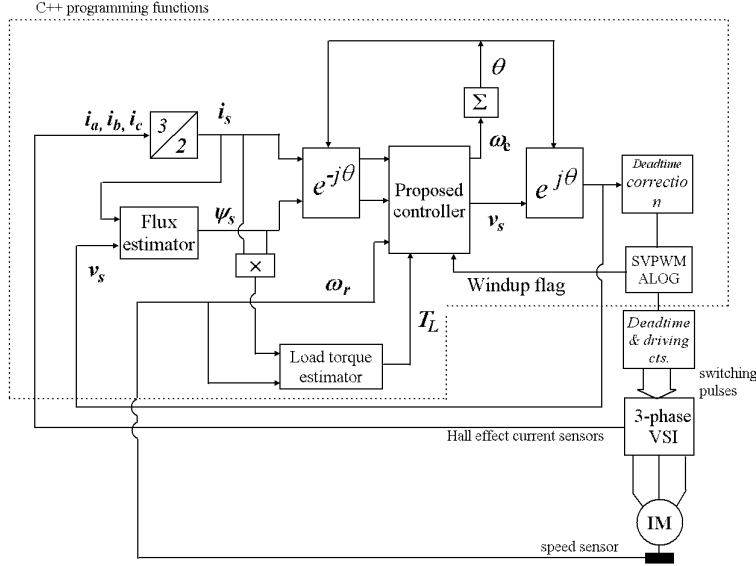


Fig. 2 Experimental system layout.

where $G = [g_1 \ g_2 \ g_3 \ g_4]$

$$F_{rj} = G_j Gr, F_{dj} = G_j G_{dk}, j = 1, 2, \dots, M$$

Feedback gain: $G = -\gamma \Gamma^T K \phi$

Feedforward gain: $G_1 = -\gamma \Gamma^T K$

$$G_2 = -\gamma \Gamma^T \phi^T \lambda$$

$$G_j = G_{j-1} K_i, j = 3, 4, \dots, M$$

$K_i = K^{-1} \phi^T \lambda$ where, K , γ and λ are the steady state solution of the following algebraic Riccati's equation

$$K(i) = Q + \Phi^T \lambda(i+1) \Phi$$

$$\lambda(i+1) = K(i+1) [I_{14} - \Gamma \gamma(i+1) \Gamma^T K(i+1)]$$

$$\gamma(i+1) = [R + \Gamma^T K(i+1) \Gamma]^{-1} \quad (14)$$

The symbol M denotes the preview feedforward steps.

The dynamic performance of the LQR-based design depends on the selection of the weighting matrices. Attempts to determine these matrices based on prespecified time or frequency domain specifications can be found, for example, in [7][9]. However, these methods imply excessive control input since they are based on the asymptotic property of the LQR (i.e., $\Re \rightarrow 0$). Therefore, repeated trial and error modifications via CAD procedure are performed, in this work, to find proper values of the weighting factors based on a compromise between the different control requirements. To facilitate this process, the weighting matrices are selected to have diagonal structure as

$$Q = \begin{bmatrix} q & 0 & 0 \\ 0 & q & 0 \\ 0 & 0 & 0 \end{bmatrix} (14 \times 14), \Re = r$$

$$q = \begin{bmatrix} q_1 & 0 & 0 \\ 0 & q_2 & 0 \\ 0 & 0 & q_3 \end{bmatrix}, R = \begin{bmatrix} r_1 & 0 & 0 \\ 0 & r_2 & 0 \\ 0 & 0 & r_3 \end{bmatrix} \quad (15)$$

V. STATOR FLUX ESTIMATOR

The stator flux linkage can be estimated in the stationary reference (d-q) frame by integrating the stator voltage minus resistive voltage drop

$$\hat{\psi}_s = \frac{v_s - R_s i_s}{s}; s = \frac{d}{dt} \quad (16)$$

The well known integration drift problem due to dc offset and measurement noise can be avoided by using low pass filter in the form [10]

$$\hat{\psi}_s = \frac{v_s - R_s i_s}{s + \omega_c} \quad (17)$$

The cut-off frequency ω_c is selected fairly small compared with the working frequency to reduce the phase and magnitude errors.

VI. COORDINATE TRANSFORMATION

The coordinate transformation angle θ is generated by integrating the synchronous speed command

$$\theta = \int \omega_e dt \quad (18)$$

with an initial value equal zero.

VII. LOAD TORQUE ESTIMATOR

The load torque, which represents the disturbance signal in (13), can be estimated from the computed electromagnetic, inertia, and friction torques as

$$\hat{T}_L = T_{em} - Js\omega_r - F\omega_r \quad (19)$$

where,

$$T_{em} = \frac{3}{4} P (\psi_{cs} i_{\beta s} - \psi_{\beta s} i_{cs}) \quad (20)$$

A variable gain digital filter can be employed at the output stage of the load torque estimator to attenuate the wide frequency range noise with minimal delay [11].

VIII. SYSTEM LAYOUT

Fig. 2 shows a schematic diagram of our implementation for the proposed control scheme. A personal computer (P-III, 750 MHz & 250 MB RAM) equipped with data acquisition and timer cards are used to realize the different control functions and data visualization. In that figure, all blocks inside the dotted line box represent software programming functions written in C++. Equations (17)-(19) are mapped into recursive digital form using trapezoidal method. In this implementation, the stator voltage is considered equal to its reference value. Therefore, a dead-time correction block is added to compensate the voltage error caused by the dead-time of the inverter [10]. The stator control voltage is transformed to the stationary frame and processed through a SVPWM algorithm with 1 kHz switching frequency. The algorithm realizes the stator control voltage by generating successive sector dependent digital words, representing the switching signals, with residence times controlled by 20 MHz clock timer [4].

The generated switching signals are then fed to a dead-time circuit where, the 3-PWM signals are complemented to get 6-PWM signals and a proper blanking time is inserted between the high and low side signals of the same phase arm. The outputs of the dead-time circuit are then isolated and amplified in the driving circuit to provide adequate gating for the 3-phase voltage source inverter. A DC bus voltage of 200 V is obtained by 3-phase diode bridge rectifier and 2000 μ F capacitor filter. Furthermore, an anti-windup flag is activated when the inverter is being saturated and this in turn will stop updating the integral part of the control law till normal operation retained again. This was important to avoid the controller windup and enhance the performance. The specifications and parameters of the IM under study are listed Appendix A. A 750W separately excited DC generator is used to provide the motor loading.

IX. SYSTEM EVALUATION

In order to achieve constant flux control and field orientation control, the desired two components of the stator flux vector $\psi_{\alpha s}$ and $\psi_{\beta s}$ (plotted in dotted lines) are set to 0.35 Wb and 0.0 Wb, respectively. The weighting factors are chosen to be $q_1=1$, $q_2=q_3=1000$, $r_1=4$, $r_2=r_3=2$ and the control sample period is set to 1 ms.

First the desired speed is ramped from 0 to 200 r/s in 0.6 s while the motor operates under a linear increasing load up to 25 % of rated torque. The desired speed is then changed abruptly with 10 r/s peak to peak during steady state conditions. The simulation and the corresponding experimental results are demonstrated in Fig. 3 and Fig. 4, respectively, with good agreement. The vertical lines depict from up to down the rotor speed, α -component of the stator flux vector, β -component of the stator flux vector, synchronous speed command, α -component of the stator voltage vector command, β -component of the stator voltage command, voltage of a-phase, current of a-phase, and, the computed voltage to frequency ratio, respectively. Fig. 5 is an enlarged part of Fig. 4 during abrupt changes in the speed command. The figures reveal that the proposed control is effective in achieving its objectives with zero steady state errors. Since the control system is independent on the value of the machine inductances, the possible saturation due to transitory variations of the stator flux during speed transients didn't show impact on the controlled drive. Fig. 6 demonstrates the ability of the drive system to track wide variation in the desired speed where, a sinusoidal speed trajectory of 440 r/s peak to peak is imposed. As shown, the system exhibits good speed tracking in both directions. The jerk in the stator current (hence torque and therefore speed) during zero crossing region is owing to the well known problem of the LPF-based stator flux estimator at low frequencies [12]. Fig. 7 shows the effect of changing the preview steps ($M=0, 2$) on the transient response of the controlled drive where, the speed command is changed from 125 r/s to 135 r/s abruptly. The figure depicts that, utilizing few future values of the desired speed can reduce the overshoot while maintaining the rise-time nearly constant. Fig. 8 shows the ability of the proposed control scheme to attenuate the

effects of sudden application and removal of 50% load torque with $M=1$.

X. CONCLUSION

A new approach to the stator field oriented control of the IM drives is developed in this paper. The scheme is based on the digital optimal state control theory with preview. The synthesized controller can achieve rotor speed control, field oriented control, and constant flux control via an optimized stator voltage command. The method is verified over wide operating range and under load torque variations through simulation and experimental studies. The effectiveness of the proposed control strategy together with its simple structure makes it attractive for the AC drive applications. Besides, the availability of a clean signal for the synchronous angular speed can be utilized to improve the performance of stator flux estimator and/or rotor speed estimator in the sensorless IM drives [13].

XI. REFERENCES

- [1] B. K. Bose, "Modern Power Electronics and AC Drives," Prentice-Hall, Inc., 2001.
- [2] X. Xu, R. DeDoncker, and D. W. Novotny, "A stator flux oriented induction machine drive," in *Proc. IEEE PESC '88*, 1988, pp. 870-876.
- [3] Osama S. Ebrahim, Mohamed F. Salem and Mohamed M. Negm, "Realization of optimal preview controller for high performance control of three phase induction machines," *MEPCON'2003*, pp 323-330, 2003.
- [4] M. M. Negm, M. F. Salem, and O. S. Ebrahim, "Design and implementation of a SVPWM technique using low cost data multifunction acquisition card," 2004 Inter. Conf. on Power System Technology, *POWERCON2004*, vol. 2, pp. 1816-1821, 2004.
- [5] M.P. Kazmierkowski and L. Maslesani, "Current control techniques for three phase voltage source PWM converters: A survey," *IEEE Trans. Ind. Elec.*, vol.45 no.5, pp 691-703, 1998.
- [6] T. Murata, T. Tsuchiya, and I. Takeda, "Vector control for induction machine on the application of optimal control," *IEEE Trans. On Ind. Electr.*, vol. 37, no. 4, pp. 283-290, 1990.
- [7] A. Matsushita, and T. Tsuchiya, "Decoupled Preview Control System and Its Application to Induction Motor Drive," *IEEE Trans. On Ind. Electr.*, vol. 42, no. 1, pp. 50-57, 1995.
- [8] M. M. M. Negm, S. A. Leithy, H. Dora, and A. Kamel, "Vector control for a three phase induction motor based on optimal regulator theory," in *Proc. IEEE Africon '96*, 1996, pp. 654-660.
- [9] T. Hiroe, T. Morimoto, S. I. Inoue, and H. Takamatsu, "A new method for selecting weight matrices of LQ regulators and its application to an industrial turbine," *Proc. 32nd Conf. on Decision and Control*, pp 3333-3334, 1993.
- [10] J. Holtz, "Sensorless control of induction machines —with or without signal injection?" *IEEE Trans. On Ind. Electr.*, Vol. 53, no. 1, pp. 7-30, 2006.
- [11] Yasuhiko Date, "Servo Motor and Motion Control Using Digital Signal Processors," Prentice-Hall, Inc., 1990.
- [12] T.W. Chun, M. K. Choi, and B. K. Bose, "A Novel Startup Scheme of Stator-Flux-Oriented Vector-Controlled Induction Motor Drive Without Torque Jerk," *IEEE Trans. On Ind. Electr.*, vol. 39, no. 3, pp. 776-782, 2003.
- [13] M. Comanescu and L. Xu, "An Improved Flux Observer Based on PLL Frequency Estimator for Sensorless Vector Control of Induction Motors," *IEEE Trans. On Ind. Electr.*, vol. 53, no. 1, pp. 50-56, 2006.

APPENDIX A

IM specifications and parameters

Specifications	4-pole, 1 kW, 130 V, 50 Hz,
R_s/R_r	0.89 / 0.92 Ω
$L_s/L_r/L_m$	0.11 / 0.11 / 0.1 H
J	0.02 N.m.s ²
F	0.0035 N.m.s.

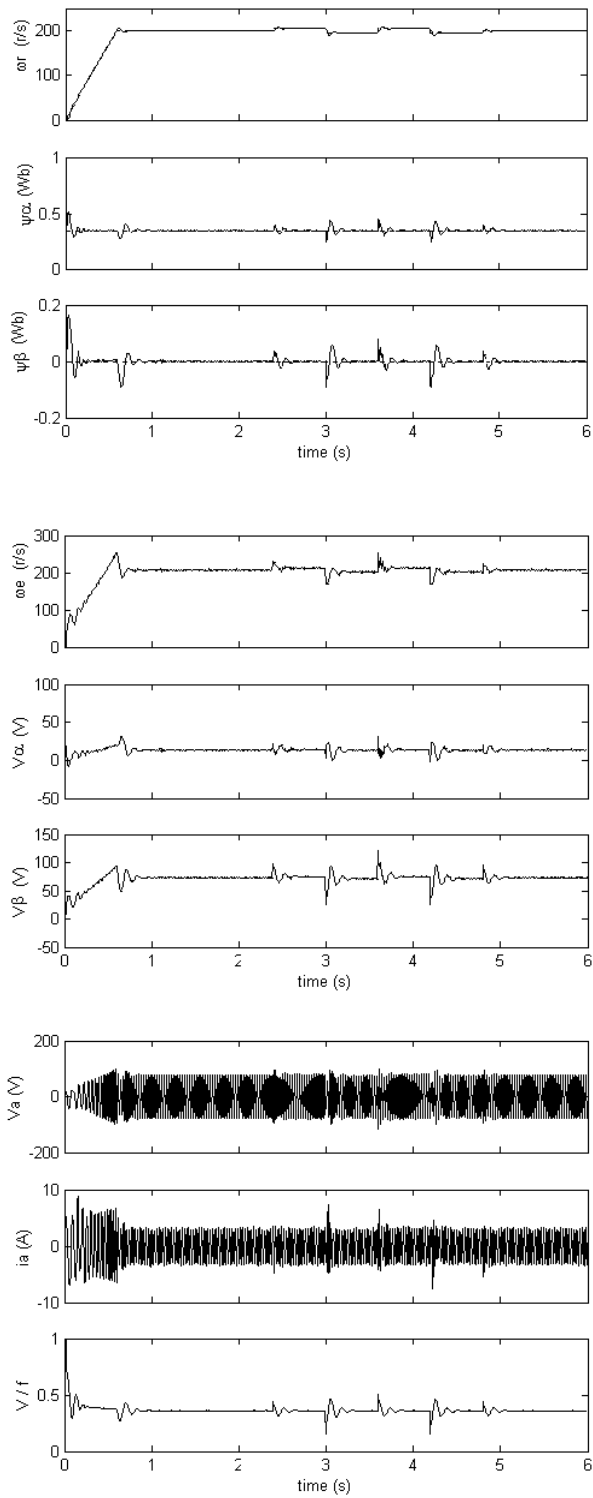


Fig. 3 Simulated drive system responses.

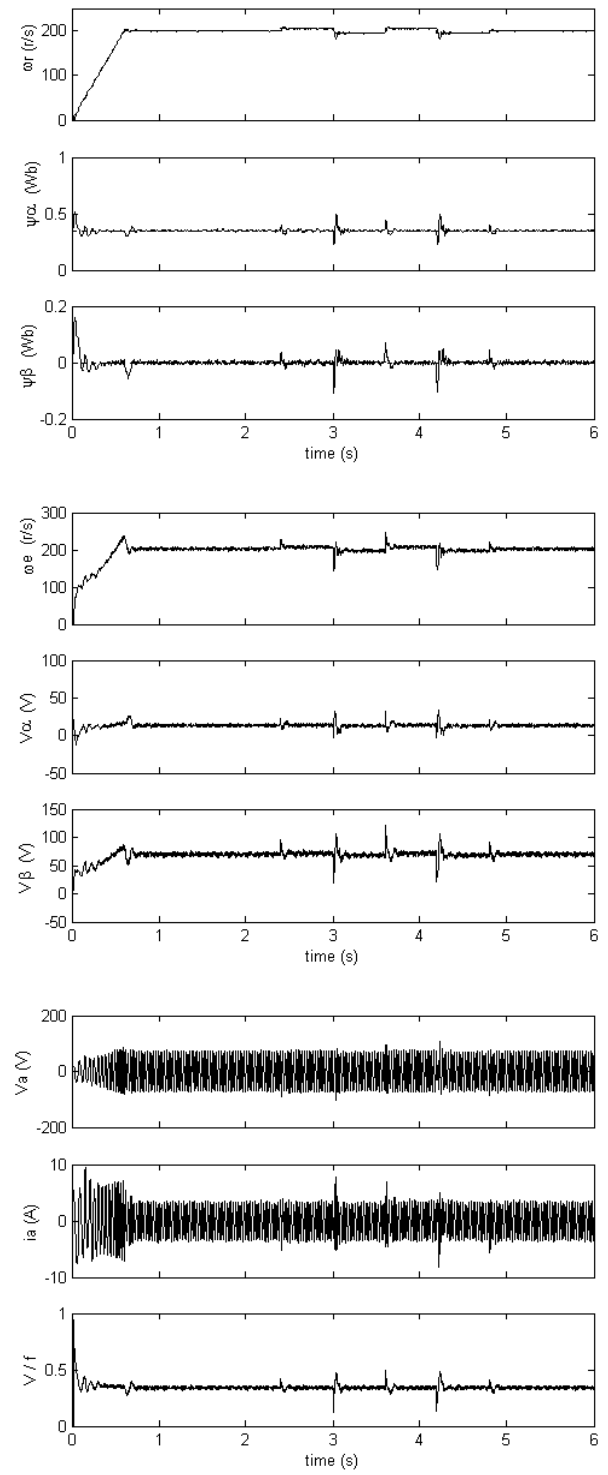


Fig. 4 Experimental drive system responses

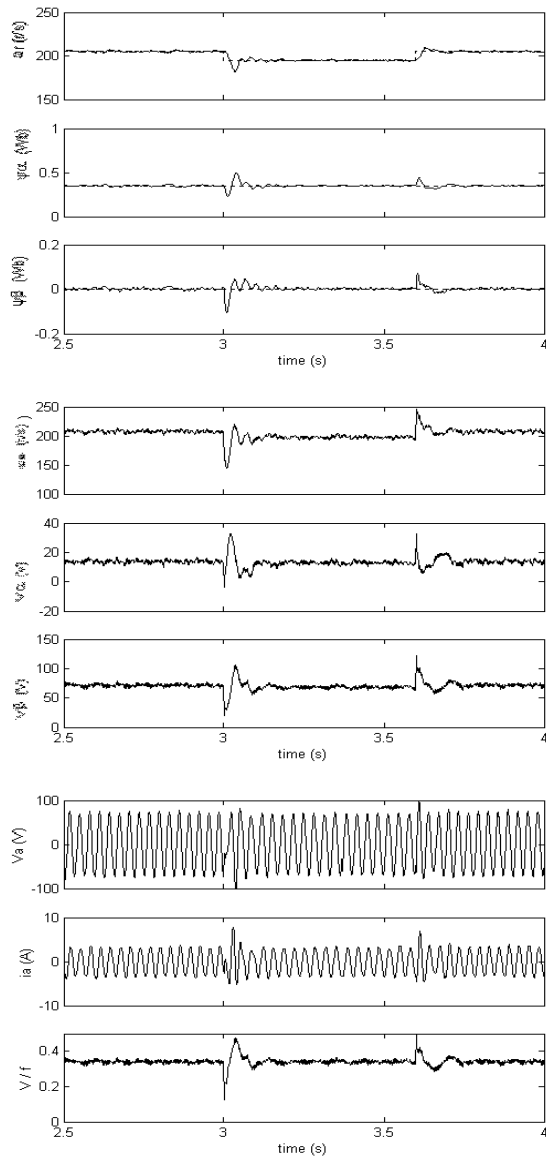


Fig. 5 An enlarged part of Fig. 4.

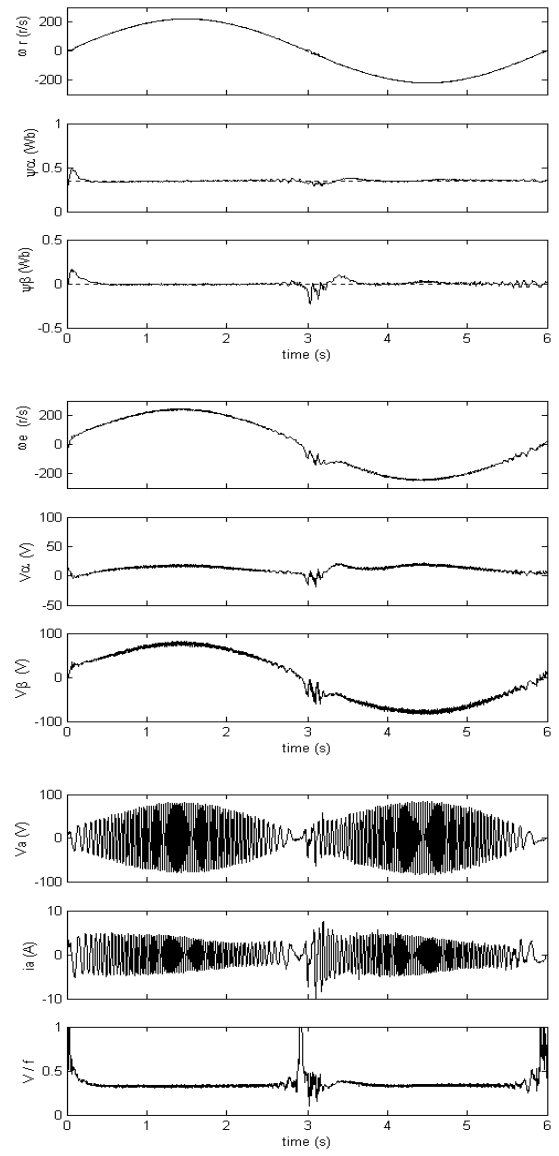


Fig. 6 Drive responses under sinusoidal speed command

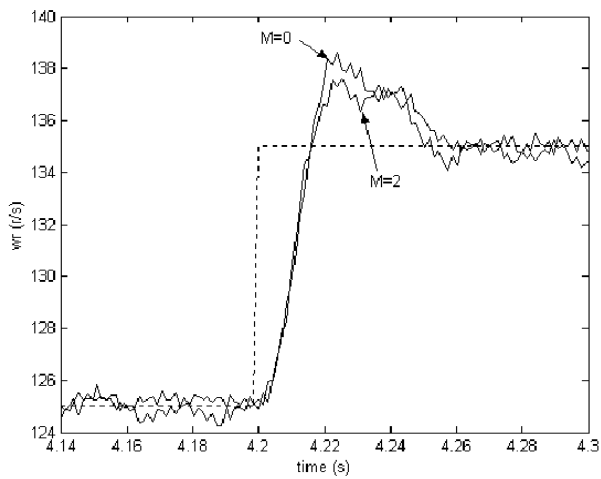


Fig. 7 Effect of changing the preview steps ($M=0,2$).

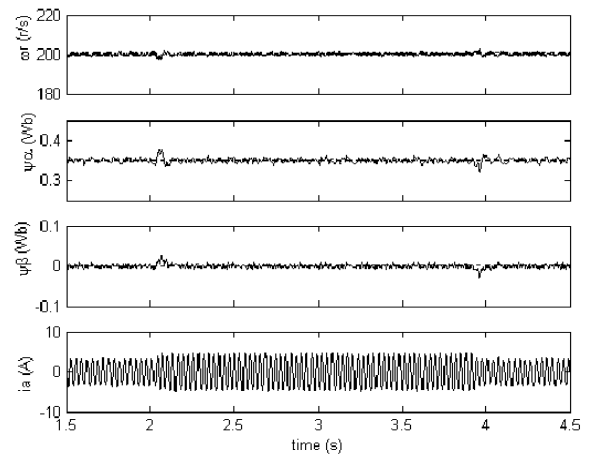


Fig. 8 Effect of changing the load torque abruptly